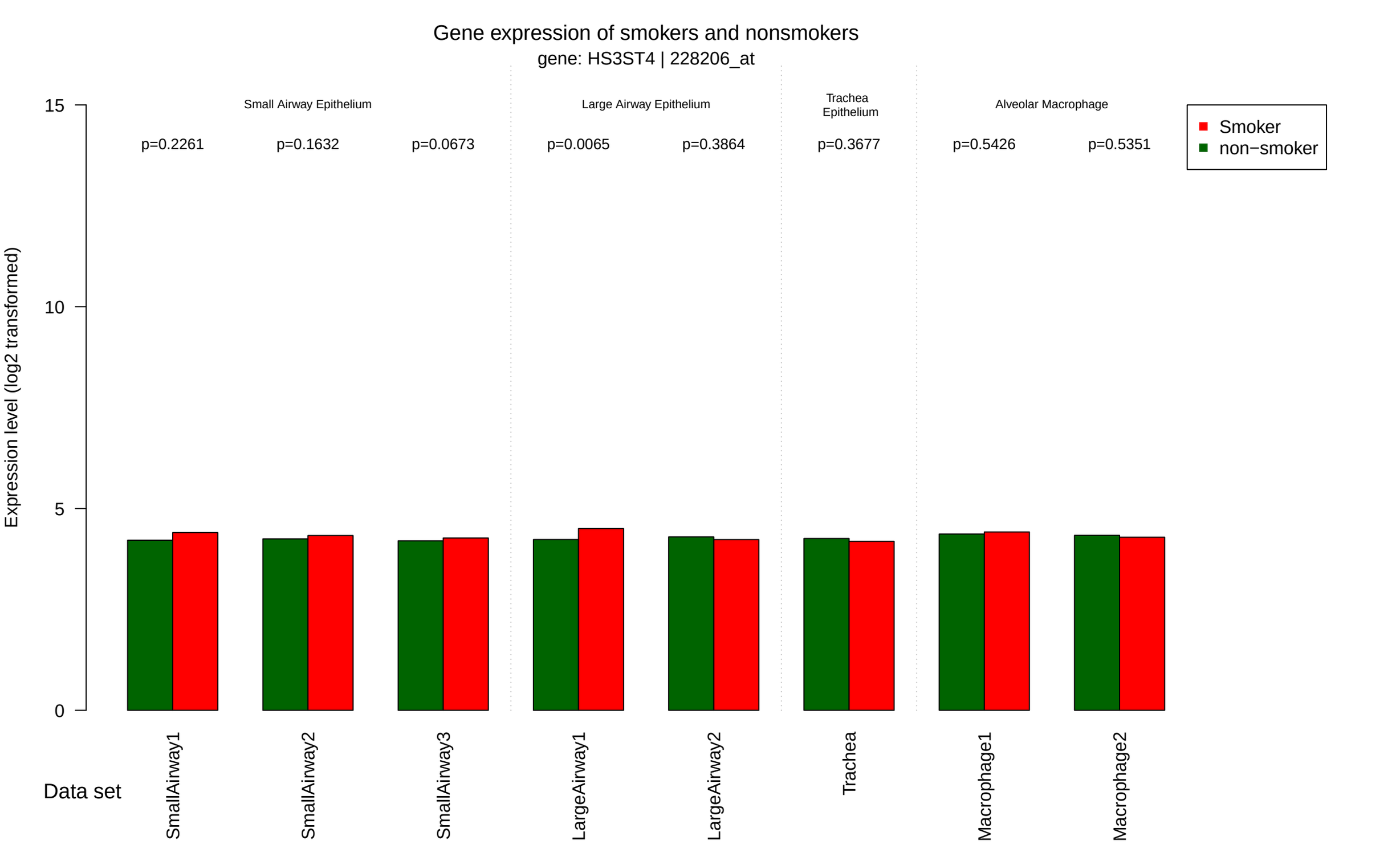
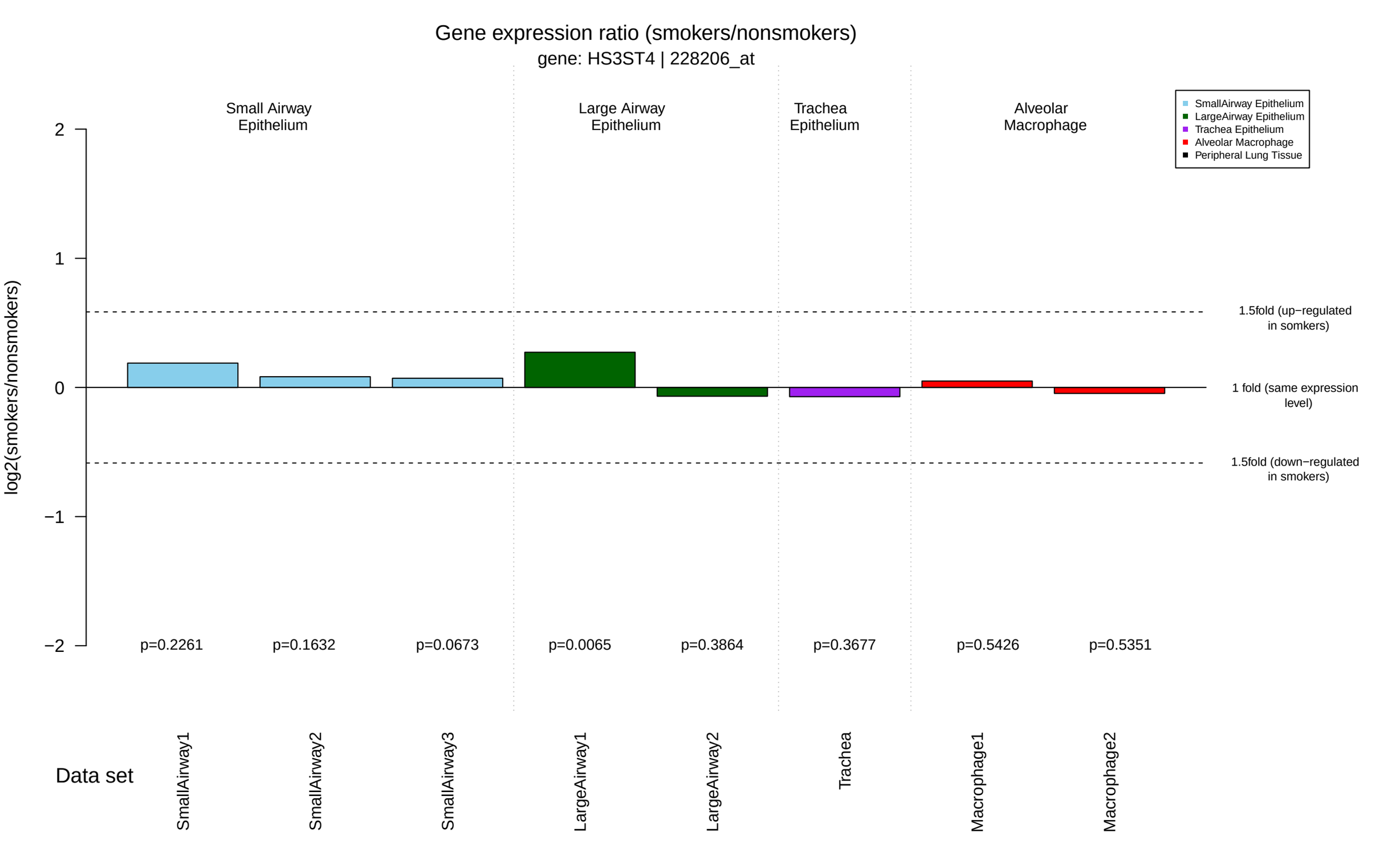
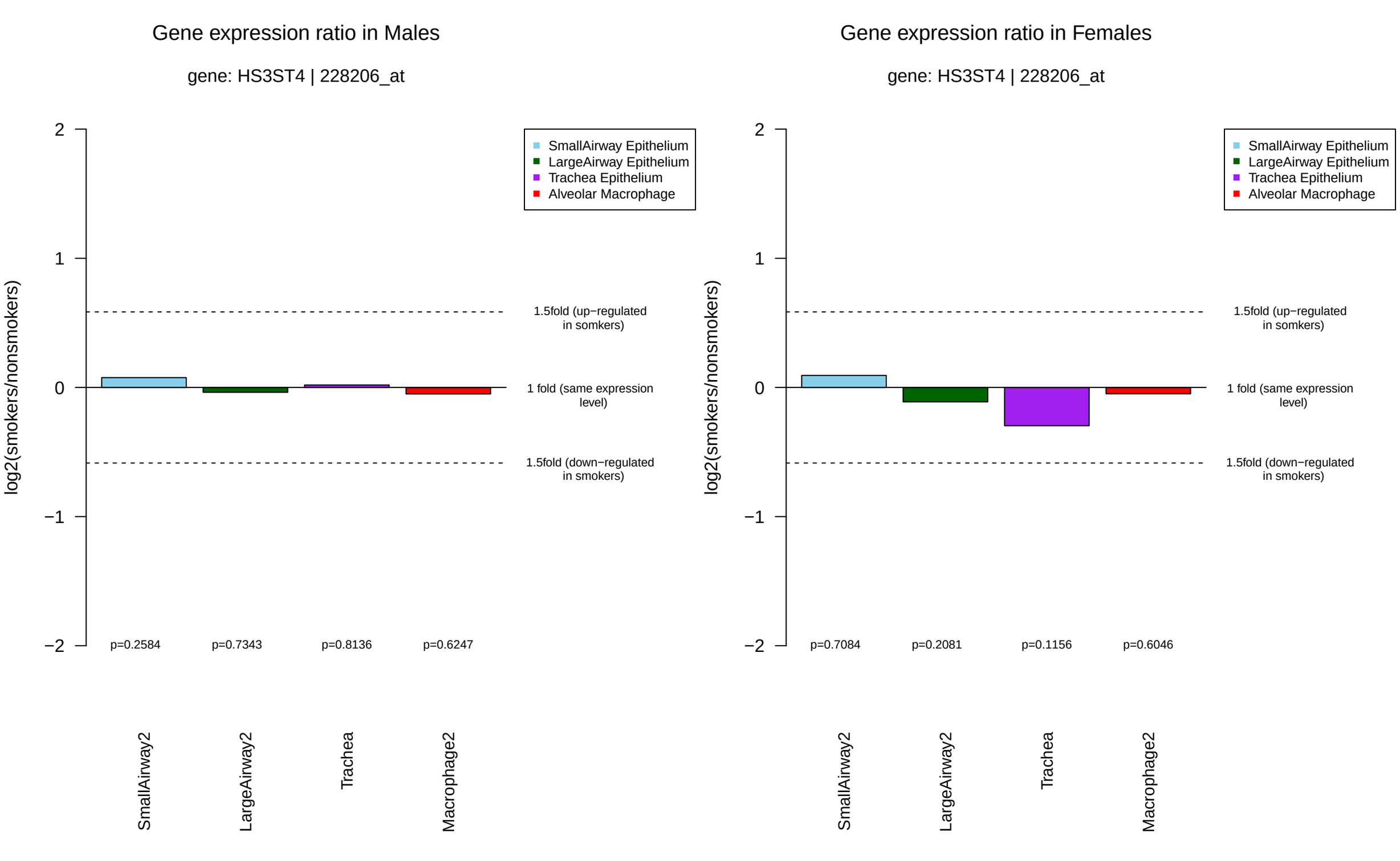
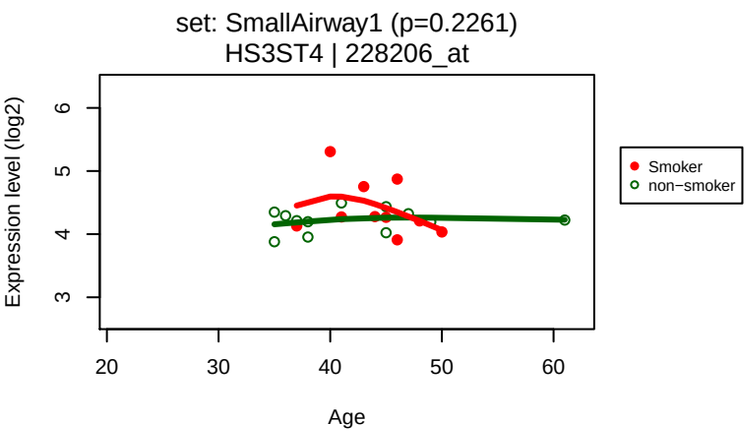
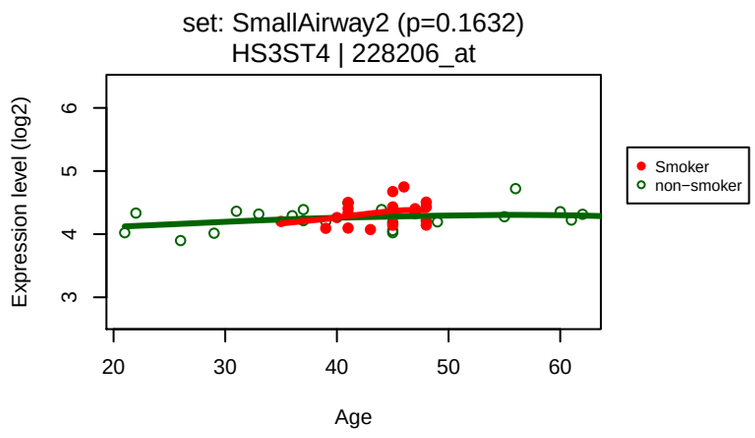
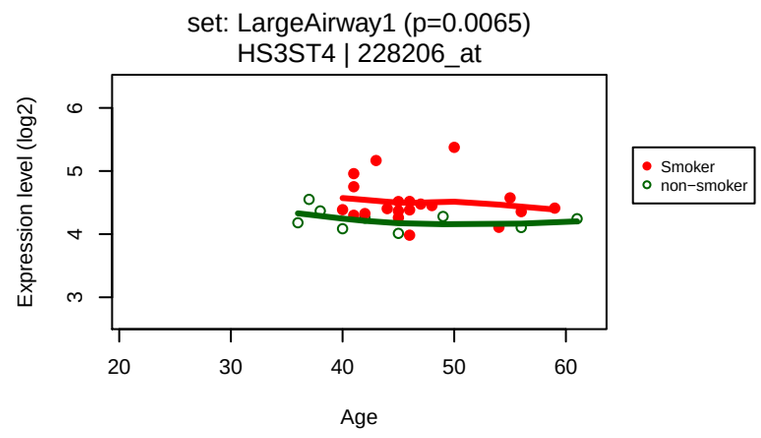

Gene expression of smokers and nonsmokers
gene: HS3ST4 | 228206_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2261
p=0.1632
p=0.0673
p=0.0065
p=0.3864
p=0.3677
p=0.5426
p=0.5351
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

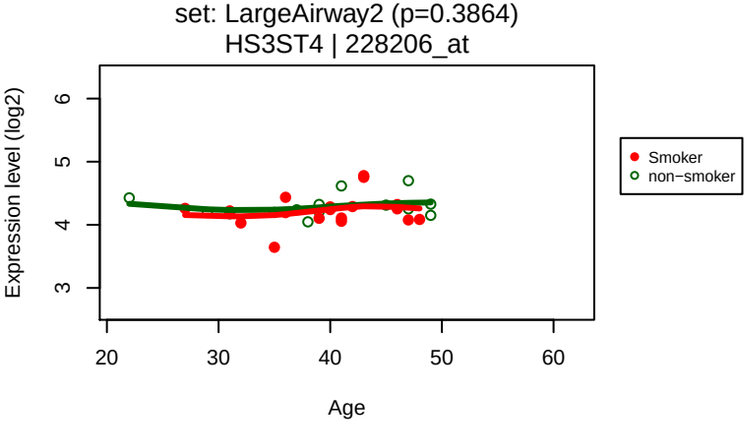

Gene expression ratio (smokers/nonsmokers)
gene: HS3ST4 | 228206_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2261
p=0.1632
p=0.0673
p=0.0065
p=0.3864
p=0.3677
p=0.5426
p=0.5351
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

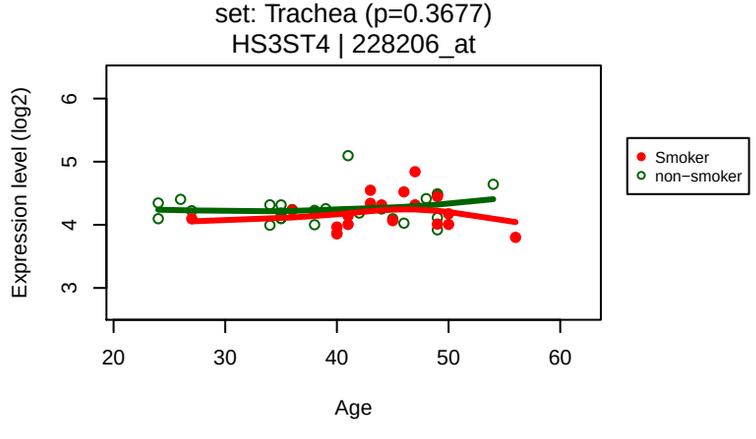

Gene expression ratio in Males
gene: HS3ST4 | 228206_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2584
p=0.7343
p=0.8136
p=0.6247
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HS3ST4 | 228206_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7084
p=0.2081
p=0.1156
p=0.6046
SmallAirway2
LargeAirway2
Trachea
Macrophage2

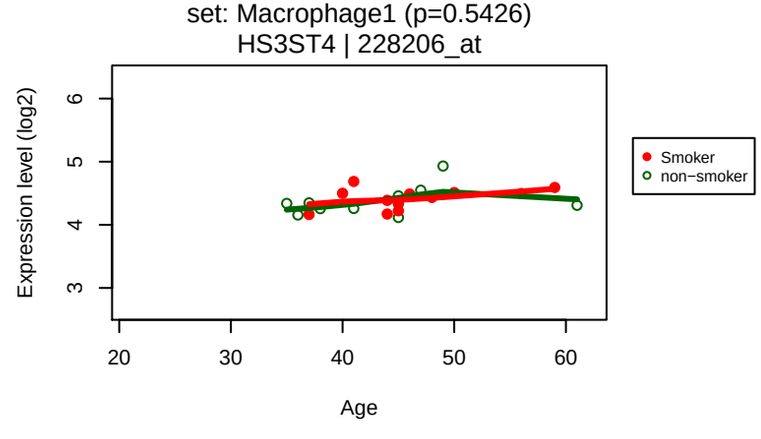

set: SmallAirway1 (p=0.2261)
HS3ST4 | 228206_at
Expression level (log2)
Age
Smoker
non−smoker

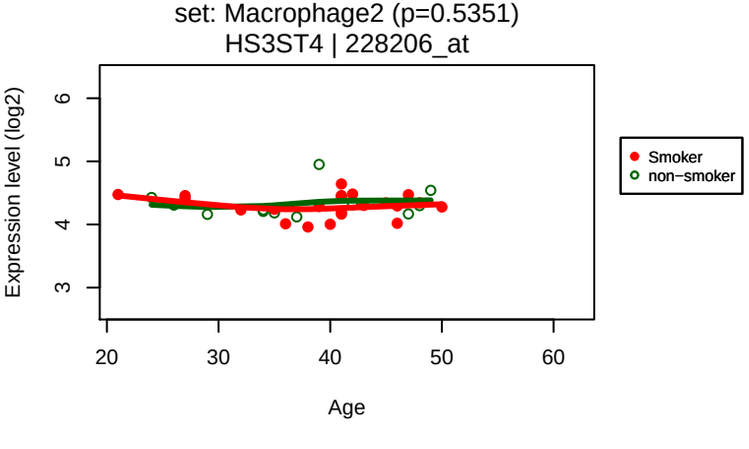

set: SmallAirway2 (p=0.1632)
HS3ST4 | 228206_at
Expression level (log2)
Age
Smoker
non−smoker

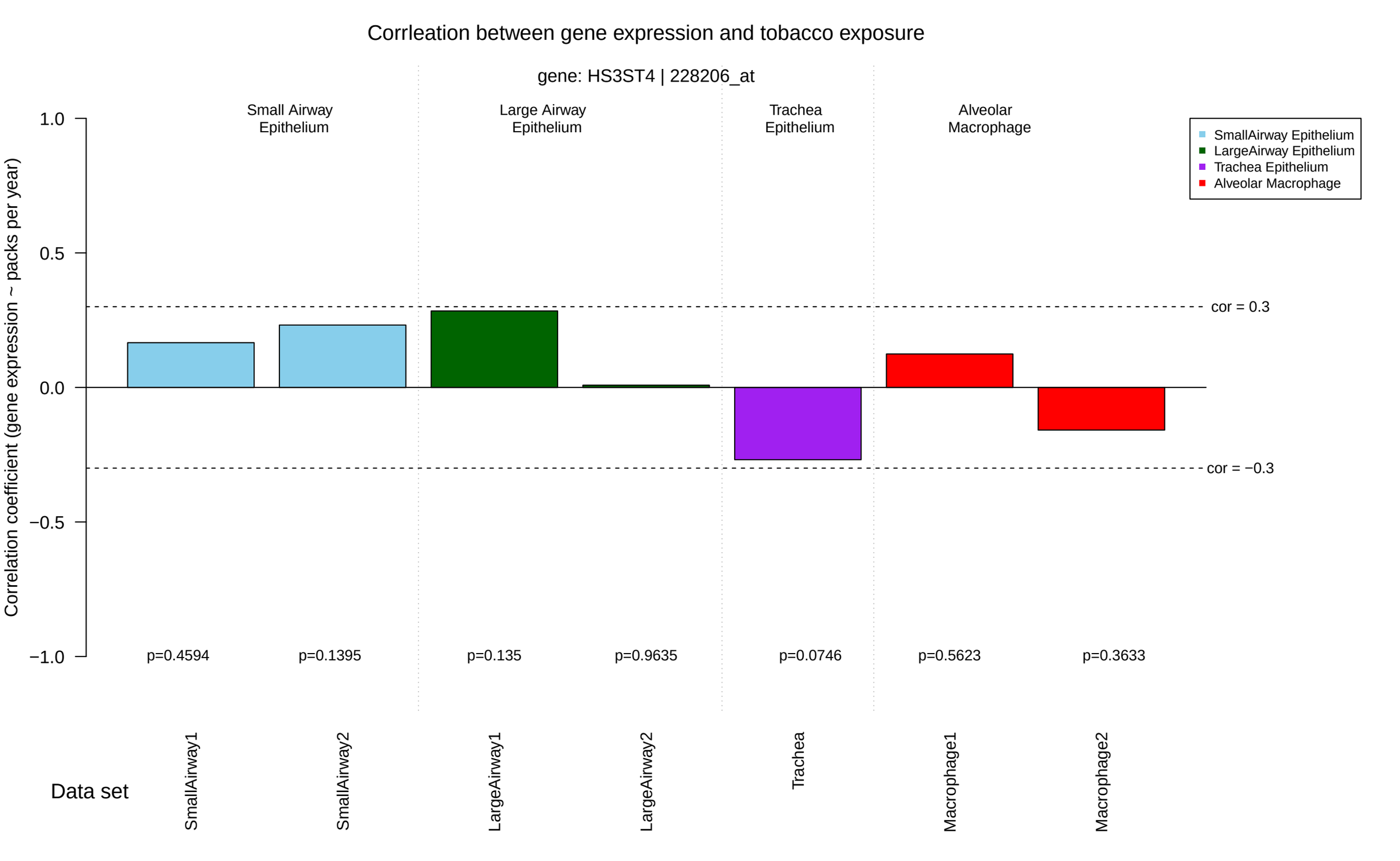

set: LargeAirway1 (p=0.0065)
HS3ST4 | 228206_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3864)
HS3ST4 | 228206_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3677)
HS3ST4 | 228206_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5426)
HS3ST4 | 228206_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5351)
HS3ST4 | 228206_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HS3ST4 | 228206_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4594
p=0.1395
p=0.135
p=0.9635
p=0.0746
p=0.5623
p=0.3633
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2